

REVIEW ARTICLE

Hormone therapy and clinical and surrogate cardiovascular endpoints in women with chronic kidney disease: a systematic review and meta-analysis

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Abstract

Objective: Women with chronic kidney disease (CKD) experience kidney dysfunction-mediated premature menopause. The role of postmenopausal hormone therapy (HT) in this population is unclear. We sought to summarize current knowledge regarding use of postmenopausal HT and cardiovascular (CV) outcomes, and established surrogate measures of CV risk in women with CKD.

Methods: This is a systematic review and meta-analysis of adult women with CKD. We searched electronic bibliographic databases (MEDLINE, EMBASE, Cochrane Central Register of Controlled Trials) (inception to 2014 December), relevant conference proceedings, tables of contents of journals, and review articles. Randomized controlled trials and observational studies examining postmenopausal HT compared with either placebo or untreated control groups were included. The intervention of interest was postmenopausal HT, and the outcome measures were all-cause and CV mortality, nonfatal CV event (myocardial infarction, stroke), and surrogate measures of CV risk (serum lipids, blood pressure).

Results: Of 12,482 references retrieved, four randomized controlled trials and two cohort studies (N = 1,666 participants) were identified. No studies reported on CV outcomes or mortality. Compared with placebo, postmenopausal HT was associated with decreased low-density lipoprotein cholesterol (−13.2 mg/dL [95% CI, −23.32 to −3.00 mg/dL]), and increased high-density lipoprotein (8.73 mg/dL [95% CI, 4.72–12.73 mg/dL]) and total cholesterol (7.96 mg/dL [95% CI, 0.07–15.84 mg/dL]). No associations were observed between postmenopausal HT triglyceride levels and blood pressure.

Conclusions: Studies examining the effect of postmenopausal HT on CV outcomes in women with CKD are lacking. Further prospective study of the role of postmenopausal HT in this high-risk group is required.

Key Words: Blood pressure – Cardiovascular disease – Chronic kidney disease – Hormone therapy – Lipids – Systematic review.

Women with chronic kidney disease (CKD) carry a high burden of all-cause and cardiovascular (CV) disease.^{1,2} Menopause at a younger age is associated with increased CV risk³ in the general population. As a

consequence of kidney-mediated hypothalamic-pituitary-gonadal dysfunction,⁴⁻⁶ the average age of menopause in women with CKD is 46.8 years, 5 years earlier than that of the general population.⁷ There is strong evidence against

Received November 4, 2015; revised and accepted February 22, 2016.

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Funding/support: S.R. is supported by the William Davies Scholarship and the Roy and Vi Baay Chair in Kidney Research. M.C.M. is supported by a Canadian Institutes of Health Research doctoral scholarship. J.M.H.-L. is funded as the Scientific Director of the Seniors Health Strategic Clinical Network within Alberta Health Services. S.B.W. is supported by a grant from St. Jude Medical and Alberta Health Services. M.T.J. is

supported by a Kidney Research Scientist Core Education and National Training new investigator award. E.W.S. is supported by a National Institute of Health grant. S.B.A. is supported by Alberta Innovates – Health Solutions. Funding sources had no role in the study design, in the writing of the report, and in the decision to submit the article for publication.

Financial disclosure/conflicts of interest: Dr Wilton reports a current grant with Arca Biopharma and past grant with Boehringer-Ingelheim. No other disclosures or conflicts of interests to report.

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treatment with hormone therapy (HT) for prevention of CV disease in postmenopausal women overall.⁸⁻¹⁰ The pathophysiology of cessation of menses in women with CKD, however, differs markedly from primary ovarian insufficiency or natural menopause,¹¹ and thus the role of HT in this population is unclear.

Recent reports have suggested that there may still be cardioprotective benefit in healthy women who initiate HT within 10 years of the onset of menopause,¹² and the initiation of estrogen therapy in young women less than 45 years with surgical menopause decreases CV mortality risk.¹³ This suggests that sex hormones could play a protective role in terms of CV risk in select populations, including women with CKD who experience early menopause. International guidelines have suggested that women with premature menopause use HT until the median age of natural menopause,¹⁴⁻¹⁷ though no guidelines exist in the case of the woman with CKD-mediated menopause. Furthermore, although large trials of healthy women have shown no CV benefit after oral HT,⁸⁻¹⁰ other important factors play a role in determining risk and benefit such as timing of initiation of HT with respect to the onset of menopause, and dose, formulation, type of progestin, and route of administration of postmenopausal HT.¹⁸⁻²⁰ These demographic and biologic differences can influence baseline CV risks and may modify the overall observed effects of HT on CV risk.

Given the potentially increased CV risk associated with low estrogen status in women with CKD and the uncertainties pertaining to the role of HT in these participants, the objective of this systematic review and meta-analysis was to determine the associations with and effects of HT on (1) all-cause and CV mortality and morbidity, and (2) surrogate measures of CV risk factors in women with CKD.

METHODS

Protocol

Our systematic review protocol was drafted using guidance from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Protocols (PRISMA-P).²¹ This protocol was registered with the international prospective systematic review register PROSPERO (CRD42014014566) and published in an open access journal.²² Our methods are described briefly here. We originally intended to examine the effect of HT on all-cause and CV mortality and morbidity in women with CKD; however, because none of the studies reported on these outcomes we expanded our outcomes of interest to include surrogate measures (lipid profile and blood pressure) of CV disease and adverse events.

Information sources and literature search

Published and unpublished studies were identified through searching electronic databases (MEDLINE, EMBASE), from inception until December 2014 without language restrictions. All relevant text words and Medical Subject Heading/Emtree terms for CKD ("Renal Insufficiency Chronic," "Kidney Failure," "Kidney Diseases," "Renal Replacement Therapy,"

"Uremia," "Dialysis," "Hemodialysis," "Peritoneal Dialysis," or "Predialysis") and HT ("Hormone Therapy," "Hormone Replacement Therapy," "Estrogen Replacement Therapy," "Estrogen," "Estradiol," "Conjugated Equine Estrogen," "Progesterone," "Progestin," "Medroxyprogesterone acetate," or "Dydrogesterone") were combined separately using the "OR" Boolean operator. These two search themes were then combined using the Boolean "AND" operator. A second search was used to identify randomized controlled trials (RCTs) studying the effect of HT on CV mortality and morbidity. In addition to the abovementioned search terms for HT, all relevant search terms for CV disease ("Cardiovascular Diseases," "Stroke," "Hypertension," "Hyperlipidemias," "Arteriosclerosis," "Cholesterol," "Blood Pressure," "Coronary Heart Disease," "Mortality," "Survival Rate," "All Cause Mortality," "Cause of Death," or "Death") and RCT were included.

The literature search was drafted by an experienced information specialist and peer reviewed by a second information specialist using the Peer Review of Electronic Search Strategies checklist.²³ This search was supplemented with a gray literature search using various methods, including scanning the reference lists of included studies, relevant reviews and published conference proceedings, and asking experts in the field.

Study selection and data extraction

After conducting a calibration exercise, we reviewed each title and abstract from the literature search independently using the inclusion and exclusion criteria. We documented the categorization of abstracts of each reviewer and quantified the chance-corrected agreement between the two reviewers. We resolved any disagreement in study inclusion by discussion and inclusion of a third reviewer (S.B.A.). The same process was followed for screening potentially relevant full-text articles. The data abstraction forms are available from the authors on request. When data were missing or clarification of published data was needed, we contacted the authors.

Inclusion criteria were as follows: (1) study population was women with CKD (as defined by authors); (2) intervention was postmenopausal HT; (3) comparator was placebo or no treatment; (4) outcome was CV mortality or morbidity, all-cause mortality, or surrogate markers of CV risk; and (5) study design was RCT or observational study.

Appraisal of methodological quality and risk of bias

Each study was appraised for quality and risk of bias by each reviewer using the Cochrane risk of bias tool for RCTs or the Cochrane risk of bias tool for cohort studies.²⁴ Study quality was assessed for different categories: for RCTs, randomization process, allocation concealment, blinding, and intention-to-treat analysis were used and for cohort studies, description of loss to follow-up, selection bias, measurement bias, and proper accounting of confounders and effect modifiers (assessed by use of cointerventions such as statins and antihypertensives) were used.

Outcome

The outcomes of interest were as follows: (1) CV mortality, all-cause mortality, and CV morbidity; (2) changes in surrogate measures of CV risk (lipid profile and blood pressure); and (3) adverse events.

Data synthesis and analysis

Summary estimates of the weighted mean differences were obtained using a DerSimonian and Laird random effects model.²⁵ The amount of variability across studies due to heterogeneity was estimated using I^2 statistic²⁶ and Cochran's Q tests of homogeneity. A two-sided $P < 0.05$ was considered statistically significant for all analyses. All analyses were performed using Stata (version 12; StataCorp LP, College Station, TX).

RESULTS

Study selection

Of 12,482 potentially relevant citations screened, 11,550 were excluded and 932 full-text publications were retrieved for further review (Fig. 1). After full-text review, 927 studies were excluded, leaving two parallel RCTs,^{27,28} one crossover RCT,⁵ and two cohort studies that met inclusion criteria.^{29,30} An additional parallel RCT was included from the gray literature search.²⁸ Chance-corrected agreement between the two independent reviewers who evaluated study eligibility was excellent with 90% agreement between the two reviewers.

Study characteristics

A total of 1,666 participants were included from the six studies (Table 1).^{5,27-29,31} All studies were designed to test the effect or association of HT in women with CKD and reported outcomes relevant to this investigation.

The mean age of study participants in the RCTs ranged from 38.5 to 62.5 years, and all were defined by the authors as postmenopausal. The mean age of the study participants in the observational studies was 51.9 years. Participants in five of the included studies (four RCTs and two cohort) had end-stage kidney disease and were on dialysis (four on hemodialysis and one unspecified),^{5,27-29,31} whereas one cohort study did not specify the stage of kidney disease.³⁰ None of the studies included participants with a kidney transplant.

None of the studies reported all-cause mortality or CV morbidity and mortality as an outcome. Among the RCTs, all four studies reported lipid levels (100%). The data were abstracted from all the studies for meta-analyses. A transdermal route of administration was examined in one RCT. The other three RCTs examined an oral route of postmenopausal hormone administration. The types of HT used in RCTs included oral estradiol,⁵ the selective estrogen-receptor modulator raloxifene,²⁷ conjugated equine estrogen,³¹ and transdermal estradiol.²⁸ In all but one RCT, which examined the effect of raloxifene,²⁷ estrogen treatment was used in combination with a progestin. The type of progestins used in RCTs included oral medroxyprogesterone acetate^{5,31} and

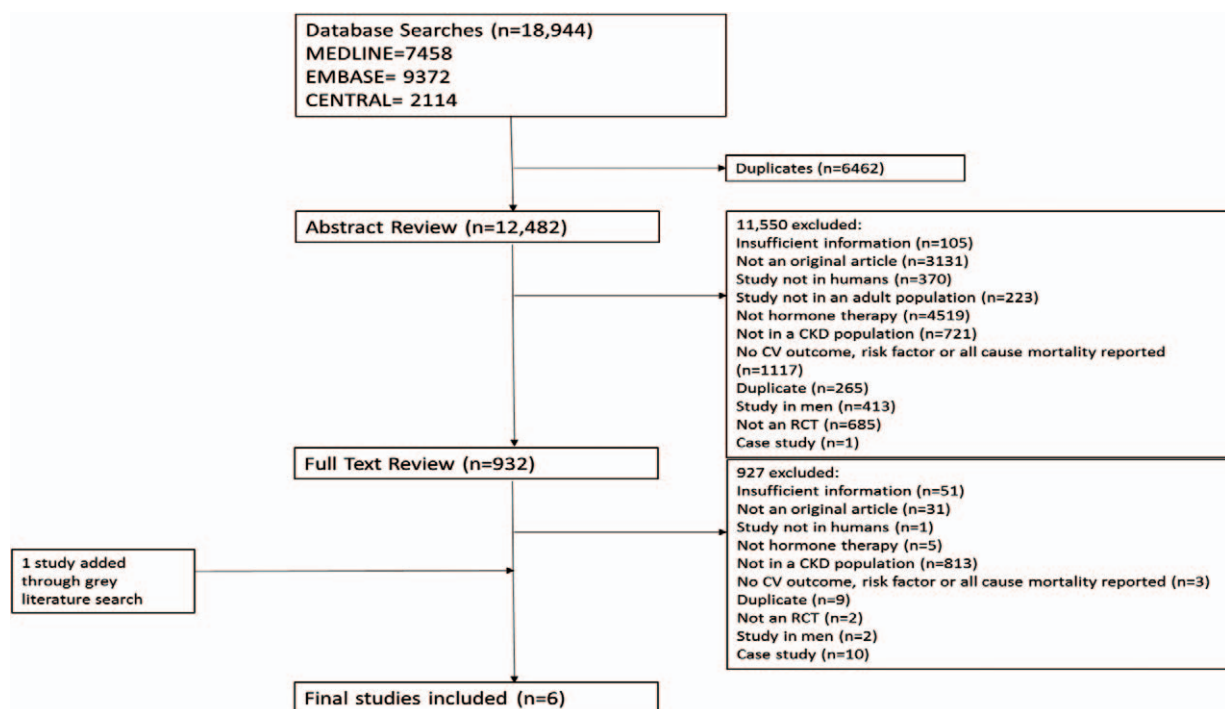


FIG. 1. PRISMA flow diagram showing the identification process for eligible studies. CKD, chronic kidney disease; CV, cardiovascular; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT, randomized controlled trial.

TABLE 1. Study characteristics

Author (date)	Study inclusion criteria	Country	Study size	Mean age in treatment	Mean age in placebo	Follow-up	Route of administration of HT/prior use of HT	Dose of HT	Study-related outcomes
Randomized controlled trials									
Ginsburg et al (1998) ⁵	Women ≥ 45 y old who were undergoing HD for ≥ 6 mo and had no menstrual periods for ≥ 1 y	United States	11	61.4	61.4	8 wk	Oral/no HT use 6 wk before study	10 mg medroxyprogesterone acetate for 1 wk followed by 1 mg estradiol for 8 wk	LDL-c, HDL-c, total cholesterol, triglycerides
Hernandez et al (2003) ²⁷	Women > 50 y old who have been menopausal for > 2 y with severe osteopenia or osteoporosis and who were treated with dialysis	Venezuela	50	63.1	61.9	52 wk	Oral/no previous use of HT	Oral raloxifene 60 mg/d	LDL-c, HDL-c, total cholesterol, triglycerides, adverse events
Matuszkiewicz-Rowinska et al (1999) ²⁸	Women who had secondary amenorrhoea, with serum estradiol levels < 30 pg/mL and were on HD for ≥ 6 mo	Poland	25	—	—	52 wk	Transdermal/no HT use 5 y before study	Transdermal estradiol 50 mg with 0.25 mg norethindrone acetate 2 times/wk	LDL-c, HDL-c, total cholesterol, triglycerides, adverse events
Park et al (2000) ³¹	Women who were on HD for ≥ 6 mo, had no menstrual periods for ≥ 1 y, and had serum estradiol levels < 25 pg/mL	Japan	65	57 (median)	61 (median)	12 wk	Oral/unknown	Oral conjugated equine estrogen 0.625 mg with medroxyprogesterone acetate 2.5 mg daily for 12 wk	LDL-c, HDL-c, triglycerides
Observational studies									
Stehman-Breen et al (1999) ²⁹ (prospective cohort)	Women in the United States Renal Data System Dialysis Morbidity and Mortality Study (DMMMS) Wave 2, which is a database of chronic hemodialysis participants	United States	1,499	—	—	Cross-sectional study	Oral/unknown	—	Total cholesterol and triglycerides
Szekacs et al (2000) ³⁰ (pre- and postintervention study)	Postmenopausal women with nephropathy and type 2 diabetes and hypertension	Hungary	16	51.9	51.9	14 wk	Oral/no previous use of HT	Oral estradiol 2 mg/d and oral norgestrel 0.5 mg/d	Blood pressure

HD, hemodialysis; HDL-c, high-density lipoprotein cholesterol; HT, hormone therapy; LDL-c, low-density lipoprotein cholesterol.

transdermal norethindrone acetate.²⁸ The duration of treatment ranged from 8 weeks to 1 year.

Both observational studies examined an oral route of administration of postmenopausal HT.^{29,30} The HT used in both studies included oral estradiol with the progestin norgestrel. One observational study reported total cholesterol and triglyceride levels,²⁹ and one observational study reported blood pressure outcome.³⁰

Study quality

The risk of bias is shown in Table 2. None of the RCTs described how the randomization was achieved or whether the investigators were aware of the allocation sequence in the study. Two of the four studies were blinded.^{5,27} All of the studies had a low risk of bias for incomplete data, selective outcome reporting, and other sources of bias. The crossover RCT included a washout period to eliminate any carryover effect.

The included observational studies had low risk of selection and measurement biases. Both studies accounted for possible confounders and effect modifiers.

Outcomes

The primary outcome of this study was to determine the effect of postmenopausal HT on all-cause and CV mortality and morbidity in women with CKD; however, none of the studies reported this outcome.

The four RCTs each reported on the effect of postmenopausal HT on low-density lipoprotein (LDL) and high-density lipoprotein (HDL) cholesterol^{5,27,28,31} in women with CKD. Neither of the observational studies reported LDL or HDL cholesterol levels.

The overall pooled estimate of the weighted mean difference in LDL cholesterol in RCTs between hormone treatment and placebo in women on hemodialysis was -13.2 mg/dL (95% CI, -23.3 to -3.00 mg/dL) or -0.34 mmol/L (95% CI, -0.60 to -0.08 mmol/L) (Fig. 2). The statistical heterogeneity among these studies was not significant ($I^2 = 7.2\%$).

The pooled estimate of the mean difference in HDL cholesterol between HT and placebo in RCTs was 8.73 mg/dL (95% CI, 0.12 - 0.33 mmol/L) (Fig. 3) or 0.23 mmol/L (95% CI, 0.12 - 0.33 mmol/L). The statistical heterogeneity among these studies was not significant ($I^2 = 14.4\%$).

Of the six studies, four RCTs and one observational study reported on total cholesterol levels and triglyceride levels. When stratified according to study design, the pooled estimate of the weighted mean difference in total cholesterol between hormone treatment and placebo was 2.94 mg/dL (95% CI, -9.90 to 15.78 mg/dL) or -0.08 mmol/L (95% CI, -0.26 to 0.41 mmol/L) in RCTs and 11.00 mg/dL (95% CI, 1.01 - 20.99 mg/dL) or 0.28 mmol/L (95% CI, 0.03 - 0.54 mmol/L) in one cohort study. The statistical heterogeneity among these studies was not significant ($I^2 < 1\%$). The pooled estimate of the weighted mean difference in total cholesterol between HT and placebo or no treatment in women with CKD was

TABLE 2. Randomized controlled trial and observational studies risk of bias assessment

Author (date)	Sequence generation	Allocation concealment	Blinding of participants, personnel, and outcome assessors	Incomplete outcome data	Selective outcome reporting	Other sources of bias
Ginsburg et al (1998) ⁵	Unclear	Unclear	Low	Low	Low	Low
Hernandez et al (2003) ²⁷	Unclear	Unclear	Low	Low	Low	Low
Matuszkiewicz-Rowinska et al (1999) ²⁸	Unclear	Unclear	Unclear	Low	Low	Low
Park et al (2000) ³¹	Unclear	Unclear	High	Low	Low	Low
Did the study match exposed and unexposed for all variables that are associated with the outcome of interest or did the statistical analysis adjust for these prognostic variables? Can we be confident that the outcome of interest was not present at start of study? Can we be confident in the assessment of the presence or absence of prognostic factors? Can we be confident in the assessment of outcome? Was the follow-up of cohorts adequate? Were cointerventions similar between groups?						
Stehman-Breen et al (1999) ²⁹	DY	DN	DY	DY	DN	PY
Szekacs et al (2000) ³⁰	DY	DY	DY	DY	PY	DY

DN, definitely no; DY, definitely yes; PN, probably no; PY, probably yes.

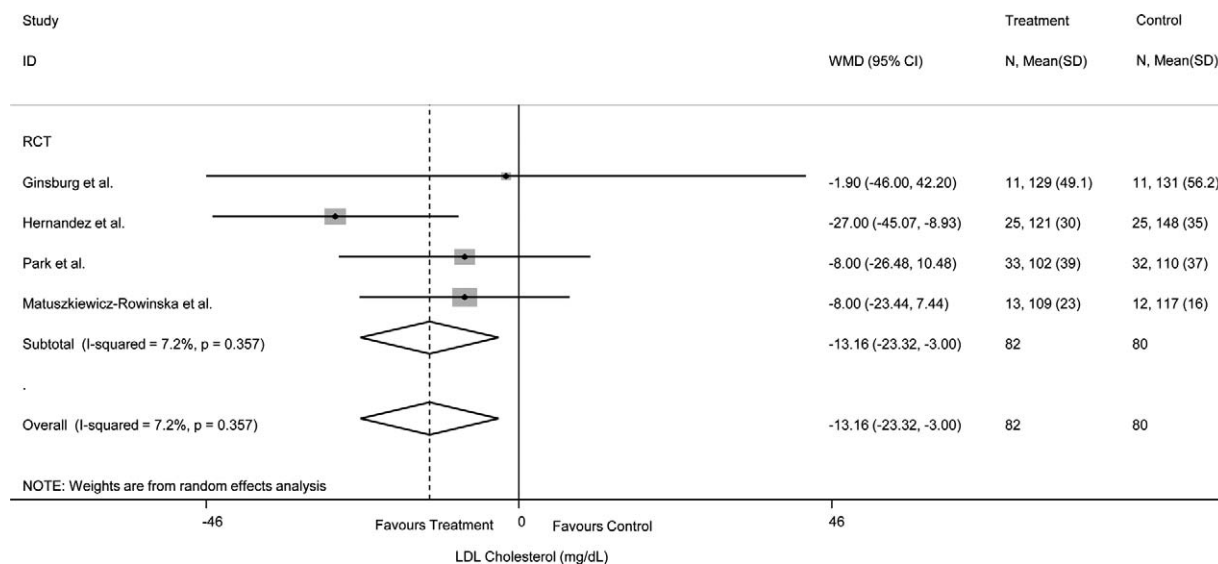


FIG. 2. Forest plot of the effect of hormone therapy on LDL cholesterol in RCTs. The WMD between treatment and placebo is presented in the figure. Study-specific WMD (95% CIs) is denoted by black dots with gray boxes (black lines). The combined WMD for all studies is represented by a black diamond, the diamond width corresponds to the 95% CI. The height of the box and the diamond is inversely proportional to the precision of the WMD estimate. LDL, low-density lipoprotein; RCT, randomized controlled trial; WMD, weighted mean difference.

7.96 mg/dL (95% CI, 0.07-15.84 mg/dL) or 0.21 mmol/L (95% CI, 0.002-0.41 mmol/L) (Fig. 4).

When stratified according to study design, the pooled estimate of the weighted mean difference in triglyceride levels between hormone treatment and placebo was -5.59 mg/dL (95% CI, -30.92 to 19.75 mg/dL) or -0.06 mmol/L (95% CI, -0.34 to 0.22 mmol/L) in RCTs and 47 mg/dL (95% CI, 21.66-72.24 mg/dL) or 0.53 mmol/L (95% CI, 0.24-0.82 mmol/L) in one cohort study. There was

significant heterogeneity among these studies ($I^2 = 74.9\%$). The pooled estimate of the mean difference in triglyceride levels between estrogen treatment and placebo or no treatment was 9.77 mg/dL (95% CI, -23.72 to 43.26 mg/dL) or 0.11 mmol/L (95% CI, -0.27 to 0.49 mmol/L) (Fig. 5).

No RCTs examined the effect of HT on blood pressure and only one of the two cohort studies investigated the effect of HT on blood pressure.³⁰ This study reported no association between oral postmenopausal HT and systolic and diastolic blood pressure.

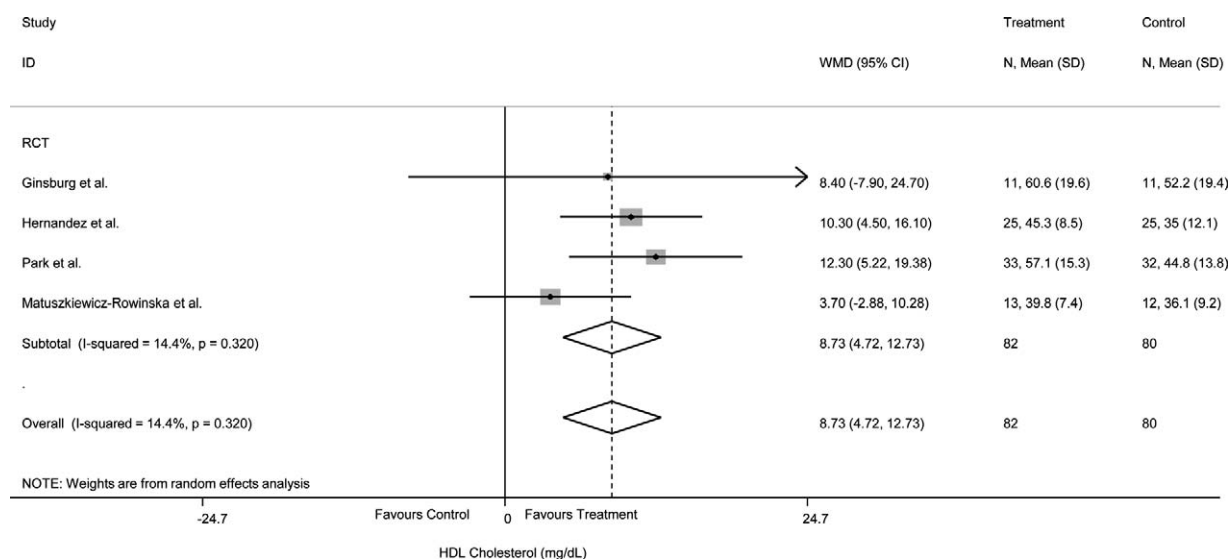


FIG. 3. Forest plot of the effect of hormone therapy on HDL cholesterol in RCTs. The WMD between treatment and placebo is presented in the figure. Study-specific WMD (95% CIs) is denoted by black dots with gray boxes (black lines). The combined WMD for all studies is represented by a black diamond, the diamond width corresponds to the 95% CI. The height of the box and the diamond is inversely proportional to the precision of the WMD estimate. HDL, high-density lipoprotein; RCT, randomized controlled trials; WMD, weighted mean difference.

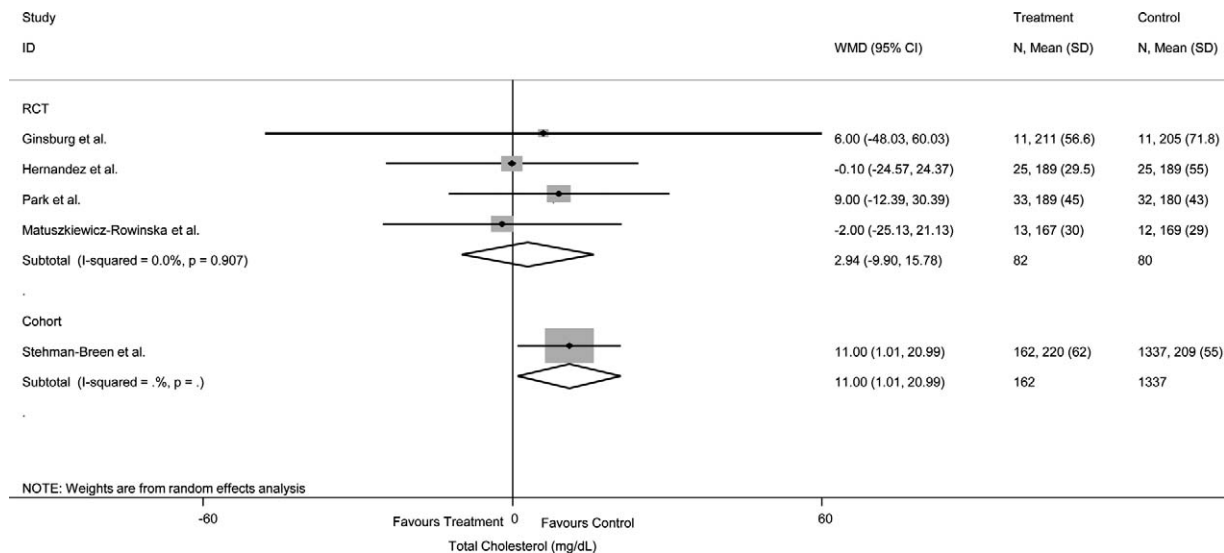


FIG. 4. Forest plot of the effect of hormone therapy on total cholesterol in RCTs and an observational study. The WMD between treatment and placebo is presented in the figure. Study-specific WMD (95% CIs) is denoted by black dots with gray boxes (black lines). The combined WMD for all RCTs and one observational study is represented by separate black diamonds, the diamond width corresponds to the 95% CI. The height of the box and the diamond is inversely proportional to the precision of the WMD estimate. RCT, randomized controlled trials; WMD, weighted mean difference.

Reporting of adverse events was inconsistent, and those reported had unclear relevance to the use of postmenopausal HT. Only two of the four RCTs and one of the two observational studies reported on adverse outcomes (n = 101 participants).^{27,28,30} In one of the RCTs two women in the treatment group developed persistent headaches and dizziness and recurrent vascular access thrombosis, but the authors did not report if this was statistically significant,²⁸ whereas in the other RCT there were no observed adverse events in either group.²⁷ No adverse outcomes were reported in the cohort study.³⁰

DISCUSSION

We performed a comprehensive review of studies examining the use of postmenopausal HT conducted among 1,600 women with CKD. Of note, the majority of women (n = 1,499) were from one prospective cohort study.²⁹ Our main findings were that in women with CKD: (1) no studies have examined the effect of postmenopausal HT on all-cause and CV mortality and morbidity; (2) HT was associated with a decrease in levels of LDL cholesterol, increases in levels of HDL cholesterol and total cholesterol, and was not associated

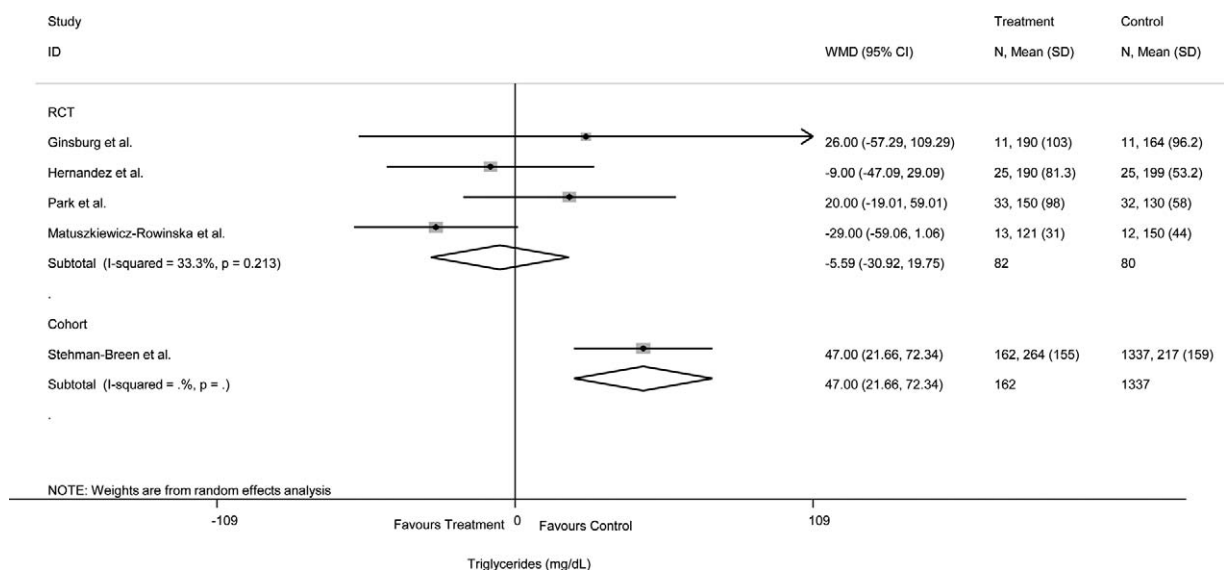


FIG. 5. Forest plot of the effect of hormone therapy on triglycerides in RCTs and an observational study. The WMD between treatment and placebo is presented in the figure. Study-specific WMD (95% CIs) are denoted by black dots with gray boxes (black lines). The combined WMD for all RCTs and one observational study is represented by separate black diamonds, the diamond width corresponds to the 95% CI. The height of the box and the diamond is inversely proportional to the precision of the WMD estimate. RCT, randomized controlled trials; WMD, weighted mean difference.

with triglyceride levels; and (3) though reporting of adverse events was limited to only three of the six studies, complications related to postmenopausal hormone treatment were not common. It is important to note, however, that the included studies had different study designs (treatment regimens, dose, and route of therapy) resulting in high heterogeneity among these studies, and this should be taken into consideration when interpreting the results.

The pathophysiology by which women with CKD develop cessation of menses differs markedly from that of natural menopause. The generally accepted definition of menopause is amenorrhea for 12 months or more and an FSH level of greater than 25 IU/L.³² Women with CKD have an abnormal sex hormone profile and abnormal hypothalamic-pituitary-gonadal activity characterized by the lack of pulsatile GnRH secretion and low estradiol,^{5,33-35} resulting in absence of menses, infertility, and premature menopause which can be reversed in the setting of kidney transplantation or quotidian hemodialysis.^{36,37} This is a distinct contrast from natural menopause whereby atresia, the continuous depletion of oocytes, ultimately results in irreversible cessation of the production of estradiol and progesterone.³⁸ Early menopause in healthy women is associated with a 2% increase in CV mortality,¹³ but the initiation of postmenopausal HT in younger women with surgical menopause³⁹ and those within 10 years of the menopausal period⁴⁰ decreases the risk (hazard ratio: 1.93 [95% CI, 1.25-2.96] in women without HT vs 1.27 [95% CI, 0.67-2.39] in women with HT). As women with CKD routinely experience menopause almost 5 years earlier than the general population,^{4,6,7,41} this suggests a potential therapeutic role for short-term estrogen treatment in this high CV risk population.⁴²

Our analysis, demonstrating a small improvement in lipid profile with hormone treatment, is similar to previous studies on this topic; however, the clinical significance of this improvement is unclear. Previous investigations in healthy women have demonstrated a favorable effect of oral estradiol on lipid profile. In the Postmenopausal Estrogen/Progestin Interventions trial of 875 healthy postmenopausal women, both oral conjugated equine estrogen only and conjugated equine estrogen plus progestin (medroxyprogesterone acetate or micronized progesterone) treatment resulted in an increase in HDL cholesterol and a decrease in LDL cholesterol.⁴³ This is further supported by studies that have shown that postmenopausal HT increases the production of HDL cholesterol and decreases clearance of HDL cholesterol.^{44,45} Caution should, however, be exercised when interpreting these results as previous studies of lipid-lowering therapy in CKD populations have not always translated into improved clinical outcomes.⁴⁶⁻⁴⁸ Of note, two studies have previously examined the role of HT on CV mortality and surrogate markers of CV disease in populations that have included women with CKD.^{49,50} In a cross-sectional analysis of 1044 postmenopausal women, a decrease in mean diastolic blood pressure was observed in HT users; however, whether this decline in blood pressure was present in participants who had CKD was

not reported.⁴⁹ Similarly, a post-hoc analysis of the Heart and Estrogen/Progestin Replacement Study trial included women with low creatinine clearance, however the reported incidence of CV disease was not stratified by treatment.⁵⁰

Although a recent meta-analysis of postmenopausal HT in healthy postmenopausal women found no evidence of cardioprotection,⁵¹ the results varied among the included studies due to the different hormone regimens, timing of HT initiation with respect to the onset of menopause, and the type of estradiol and progestin used.^{51,52} Furthermore, 8 of the 13 trials included in the meta-analysis were examining the effect of HT as secondary, rather than primary, prevention in a high CV risk population.⁵¹

There are limitations to this systematic review. First, baseline characteristics of participants varied with respect to age, type of HT, and route of administration and the paucity of studies did not allow for analyses stratified by these factors. Second, there is evidence that serum estradiol levels are variable, depending on the route of administration⁵³ and degree of CKD.¹¹ It is important to note that most of the participants in the included studies in this systematic review were women with end-stage kidney disease treated with dialysis. As there is evidence to suggest that hormonal irregularities are seen as early as stage 3 in women with nondialysis-dependent CKD,¹¹ the effect of postmenopausal HT at an earlier stage of CKD may differ compared with later stages of CKD. Our results, however, suggest that at least in the population with end-stage kidney disease treated with dialysis, HT is associated with an improved lipid profile. Third, all studies included a modest number of participants with short duration of postmenopausal hormone treatment and reported surrogate markers of CV risk rather than clinical outcomes. Fourth, reporting of adverse events was limited across studies, and most studies were not powered sufficiently to determine the rate of adverse events associated with the use of HT in this population. Moreover, the incidence of malignancy is higher in participants with end-stage kidney disease on dialysis⁵⁴; as postmenopausal HT has been associated with an increase in breast cancer risk,⁴⁰ this would be an important consideration with the use of this therapy in the CKD population.

CONCLUSIONS

In conclusion, this meta-analysis suggests potential short-term efficacy of postmenopausal HT in improving lipid profiles in younger women with CKD. No conclusions, however, can be drawn related to CV morbidity or mortality. The North American Menopause Society endorses the use of HT for short-term symptom management in women with symptoms of menopause, and highlights the need for further research with respect to the long-term outcomes of HT.⁵⁵ Currently, there are no Kidney Disease Improving Global Outcomes guidelines on the use of HT in women with CKD. A large, prospective trial in women with variable hormone status and stages of CKD, which includes clinical outcomes, is needed to guide clinical practice in terms of the role for postmenopausal HT among this high-risk population.

Acknowledgments: We thank Ms Diane Lorenzetti and Ms Helen Lee Robertson for conducting the literature searches and peer reviewing the MEDLINE search strategy.

REFERENCES

- Coresh J, Selvin E, Stevens LA, et al. Prevalence of chronic kidney disease in the United States. *JAMA* 2007;298:2038-2047.
- Go AS, Chertow GM, Fan D, McCulloch CE, Hsu CY. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med* 2004;351:1296-1305.
- Kannel WB, Hjortland MC, McNamara PM, Gordon T. Menopause and risk of cardiovascular disease: the Framingham study. *Ann Intern Med* 1976;85:447-452.
- Holley JL, Schmidt RJ, Bender FH, Dumler F, Schiff M. Gynecologic and reproductive issues in women on dialysis. *Am J Kidney Dis* 1997;29:685-690.
- Ginsburg ES, Walsh B, Greenberg L, Price D, Chertow GM, Owen WF. Effects of estrogen replacement therapy on the lipoprotein profile in postmenopausal women with ESRD. *Kidney Int* 1998;54:1344-1350.
- Kramer HM, Curhan GC, Singh A. Permanent cessation of menses and postmenopausal hormone use in dialysis-dependent women: the HELP study. *Am J Kidney Dis* 2003;41:643-650.
- Kramer HM, Curhan G, Singh A. Hemodialysis and estrogen levels in postmenopausal (HELP) patients: the multicenter HELP study. *Am J Kidney Dis* 2003;41:1240-1246.
- Rossouw JE, Anderson GL, Prentice RL, et al. Risks and benefits of estrogen plus progestin in healthy postmenopausal women: principal results from the Women's Health Initiative randomized controlled trial. *JAMA* 2002;288:321-333.
- Anderson GL, Limacher M, Assaf AR, et al. Effects of conjugated equine estrogen in postmenopausal women with hysterectomy: the Women's Health Initiative randomized controlled trial. *JAMA* 2004;291:1701-1712.
- Hulley S, Grady D, Bush T, et al. Randomized trial of estrogen plus progestin for secondary prevention of coronary heart disease in postmenopausal women. Heart and Estrogen/progestin Replacement Study (HERS) Research Group. *JAMA* 1998;280:605-613.
- Rathi M, Ramachandran R. Sexual and gonadal dysfunction in chronic kidney disease: pathophysiology. *Indian J Endocr Metab* 2012;16:214-219.
- Boardman HM, Hartley L, Eisinga A, et al. Hormone therapy for preventing cardiovascular disease in post-menopausal women. *Cochrane Database Syst Rev* 2015;3:CD002229.
- Osssewarde ME, Bots ML, Verbeek AL, et al. Age at menopause, cause-specific mortality and total life expectancy. *Epidemiology* 2005;16:556-562.
- De Villiers T, Pines A, Panay N, et al. Updated 2013 International Menopause Society recommendations on menopausal hormone therapy and preventive strategies for midlife health. *Climacteric* 2013;16:316-337.
- Shifren JL, Gass ML; NAMS Recommendations for Clinical Care of Midlife Women Working Group. The North American Menopause Society recommendations for clinical care of midlife women. *Menopause* 2014;21:1038-1062.
- Panay N, Hamoda H, Arya R, et al. The 2013 British Menopause Society & Women's Health Concern recommendations on hormone replacement therapy. *Menopause Int* 2013;19:59-68.
- The Royal Australian and New Zealand College of Obstetricians and Gynaecologists. Management of the menopause 2014. Available at: <https://www.ranzcog.edu.au/document-library/hormone-replacement-the-rapy-advice.html>. Accessed January 1, 2016.
- Manson JE, Chlebowski RT, Stefanick ML, et al. Menopausal hormone therapy and health outcomes during the intervention and extended post-stopping phases of the Women's Health Initiative randomized trials. *JAMA* 2013;310:1353-1368.
- Shufelt CL, Merz CN, Prentice RL, et al. Hormone therapy dose, formulation, route of delivery, and risk of cardiovascular events in women: findings from the Women's Health Initiative Observational Study. *Menopause* 2014;21:260-266.
- Grodstein F, Clarkson TB, Manson JE. Understanding the divergent data on postmenopausal hormone therapy. *N Engl J Med* 2003;348:645-650.
- Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1.
- Ramesh S, Mann MC, Holroyd-Leduc JM, et al. The effect of hormone therapy on all-cause and cardiovascular mortality in women with chronic kidney disease: protocol for a systematic review and meta-analysis. *Syst Rev* 2015;4:44.
- Sampson M, McGowan J, Cogo E, Grimshaw J, Moher D, Lefebvre C. An evidence-based practice guideline for the peer review of electronic search strategies. *J Clin Epidemiol* 2009;62:944-952.
- Higgins JP, Altman DG. Assessing risk of bias in included studies. In: Higgins JP, Green S, editors. *Cochrane Handbook for Systematic Reviews of Interventions: Cochrane Book Series*. Chichester, UK: John Wiley & Sons; 2008:187-241.
- DerSimonian R, Kacker R. Random-effects model for meta-analysis of clinical trials: an update. *Contemp Clin Trials* 2007;28:105-114.
- Deeks JJ, Higgins J, Altman DG. Analysing data and undertaking metaanalyses. In: Higgins JP, Green S, eds. *Cochrane Handbook for Systematic Reviews of Interventions: Cochrane Book Series*. Chichester, UK: John Wiley & Sons; 2008:243-296.
- Hernandez E, Valera R, Alonzo E, et al. Effects of raloxifene on bone metabolism and serum lipids in postmenopausal women on chronic hemodialysis. *Kidney Int* 2003;63:2269-2274.
- Matuszkiewicz-Rowinska J, Skorzewska K, Radowicki S, et al. The benefits of hormone replacement therapy in pre-menopausal women with oestrogen deficiency on haemodialysis. *Nephrol Dial Transplant* 1999;14:1238-1243.
- Stehman-Breen CO, Gillen D, Gipson D. Prescription of hormone replacement therapy in postmenopausal women with renal failure. *Kidney Int* 1999;56:2243-2247.
- Szekacs B, Vajo Z, Varbiro S, et al. Postmenopausal hormone replacement improves proteinuria and impaired creatinine clearance in type 2 diabetes mellitus and hypertension. *BJOG* 2000;107:1017-1021.
- Park JS, Jung HH, Yang WS, Kim SB, Min WK, Chi HS. Effects of hormonal replacement therapy on lipid and haemostatic factors in postmenopausal ESRD patients. *Nephrol Dial Transplant* 2000;15:1835-1840.
- Hale GE, Zhao X, Hughes CL, Burger HG, Robertson DM, Fraser IS. Endocrine features of menstrual cycles in middle and late reproductive age and the menopausal transition classified according to the Staging of Reproductive Aging Workshop (STRAW) staging system. *J Clin Endocrinol Metab* 2007;92:3060-3067.
- Veldhuis JD, Johnson ML, Bolton WK. Analyzing pulsatile endocrine data in patients with chronic renal failure: a brief review of deconvolution techniques. *Pediatr Nephrol* 1991;5:522-528.
- Swamy AP, Woolf PD, Cestero RV. Hypothalamic-pituitary-ovarian axis in uremic women. *J Lab Clin Med* 1979;93:1066-1072.
- Lim VS, Henriquez C, Sievertsen G, Frohman LA. Ovarian function in chronic renal failure: evidence suggesting hypothalamic anovulation. *Ann Intern Med* 1980;93:21-27.
- Barua M, Hladunewich M, Keunen J, et al. Successful pregnancies on nocturnal home hemodialysis. *Clin J Am Soc Nephrol* 2008;3:392-396.
- McKay DB, Josephson MA. Pregnancy after kidney transplantation. *Clin J Am Soc Nephrol* 2008;3(suppl 2):S117-S125.
- Faddy M, Gosden R, Gougeon A, Richardson SJ, Nelson J. Accelerated disappearance of ovarian follicles in mid-life: implications for forecasting menopause. *Hum Reprod* 1992;7:1342-1346.
- Rocca WA, Grossardt BR, de Andrade M, Malkasian GD, Melton LJ III. Survival patterns after oophorectomy in premenopausal women: a population-based cohort study. *Lancet Oncol* 2006;7:821-828.
- Rossouw JE, Prentice RL, Manson JE, et al. Postmenopausal hormone therapy and risk of cardiovascular disease by age and years since menopause. *JAMA* 2007;297:1465-1477.
- McKinlay SM, Brambilla DJ, Posner JG. The normal menopause transition. *Maturitas* 2008;61:4-16.
- Collins AJ, Li S, Ma JZ, Herzog C. Cardiovascular disease in end-stage renal disease patients. *Am J Kidney Dis* 2001;38:S26-S29.
- Miller VT, LaRosa J, Barnabei V, et al. Effects of estrogen or estrogen/progestin regimens on heart disease risk factors in postmenopausal women: the postmenopausal estrogen/progestin interventions (PEPI) trial. *JAMA* 1995;273:199-208.
- Walsh BW, Schiff I, Rosner B, Greenberg L, Ravnkar V, Sacks FM. Effects of postmenopausal estrogen replacement on the concentrations and metabolism of plasma lipoproteins. *N Engl J Med* 1991;325:1196-1204.
- Tikkanen MJ, Nikkilä EA, Kuusi T, Sipinen S. High density lipoprotein-2 and hepatic lipase: reciprocal changes produced by estrogen and norgestrel. *J Clin Endocrinol Metab* 1982;54:1113-1117.

46. Holdaas H, Holme I, Schmieder RE, et al. Rosuvastatin in diabetic hemodialysis patients. *J Am Soc Nephrol* 2011;22:1335-1341.
47. Krane V, Winkler K, Drechsler C, Lilienthal J, März W, Wanner C. Effect of atorvastatin on inflammation and outcome in patients with type 2 diabetes mellitus on hemodialysis. *Kidney Int* 2008;74:1461-1467.
48. Baigent C, Landray MJ, Reith C, et al. The effects of lowering LDL cholesterol with simvastatin plus ezetimibe in patients with chronic kidney disease (Study of Heart and Renal Protection): a randomised placebo-controlled trial. *Lancet* 2011;377:2181-2192.
49. Fung MM, Poddar S, Bettencourt R, Jassal SK, Barrett-Connor E. A cross-sectional and 10-year prospective study of postmenopausal estrogen therapy and blood pressure, renal function, and albuminuria: the Rancho Bernardo Study. *Menopause* 2011;18:629-637.
50. O'Hare AM, Vittinghoff E, Hsia J, Shlipak MG. Renal insufficiency and the risk of lower extremity peripheral arterial disease: results from the Heart and Estrogen/Progestin Replacement Study (HERS). *J Am Soc Nephrol* 2004;15:1046-1051.
51. Main C, Knight B, Moxham T, et al. Hormone therapy for preventing cardiovascular disease in post-menopausal women. *Cochrane Database Syst Rev* 2013;4:CD002229.
52. Grodstein F, Clarkson TB, Manson JE. Understanding the divergent data on postmenopausal hormone therapy. *N Engl J Med* 2003;348:645-650.
53. Scott RT Jr, Ross B, Anderson C, Archer DF. Pharmacokinetics of percutaneous estradiol: a crossover study using a gel and a transdermal system in comparison with oral micronized estradiol. *Obstet Gynecol* 1991;77:758-764.
54. Maisonneuve P, Agodoa L, Gellert R, et al. Cancer in patients on dialysis for end-stage renal disease: an international collaborative study. *Lancet* 1999;354:93-99.
55. North American Menopause Society. The 2012 hormone therapy position statement of The North American Menopause Society. *Menopause* 2012;19:257-271.